

KINETICS OF RECTILINEAR TRANSLATION

Principles of dynamics

1. Newton's Ist law

A body originally at rest, or moving in a straight line with a constant velocity, will remain in this state provided that it is not subjected to an unbalanced force. This defines the property of inertia of a body.

2. Newton's IInd law

A particle acted upon by an unbalanced force F experiences an acceleration a that has the same direction as the force and a magnitude proportional to the force.

Measurements of force and acceleration can be done in a laboratory. If a known force F_1 is applied to a particle, the acceleration a_1 can be measured. The ratio of $\frac{F_1}{a_1}$ is found out. A different force F_2 will create a different acceleration a_2 from which the ratio $\frac{F_2}{a_2}$ is evaluated. Both the ratios will be the same.

$$\frac{F_1}{a_1} = \frac{F_2}{a_2} = m,$$

the constant is called the mass of the particle. The mass m provides a quantitative measure of the resistance of the particle to a change in its velocity.

The second law can be written as $F = ma$, where F is the resultant force acting on the body. This is referred to as the **Equation of motion**.

3. Newton's IIIrd law

The mutual forces of action and reaction between two particles are equal, opposite, and collinear.

Differential equation of rectilinear translation

When more than one force acts on a particle, the resultant force is determined from ΣF_x and ΣF_y . If $\Sigma F_y = 0$ and $\Sigma F_x \neq 0$, the body will move in the x-direction. Then the equation of motion can be written as $\Sigma F_x = ma$. Since $a = \frac{d^2x}{dt^2}$. The equation motion is

$$m \frac{d^2x}{dt^2} = \Sigma F_x \quad \text{or} \quad m\ddot{x} = \Sigma F_x$$

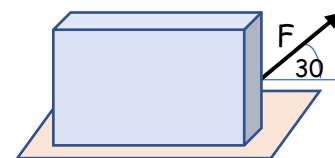
This is the differential equation of rectilinear motion.

Inertial frame of reference

Whenever the equation of motion is applied, it is required that measurements of the acceleration be made from a Newtonian or inertial frame of reference. Such a co-ordinate system does not rotate and is either fixed or translates in a given direction with a constant velocity (zero acceleration). This ensures that the particle's acceleration measured by observers in two different inertial frames of reference will always be the same.

Example 1

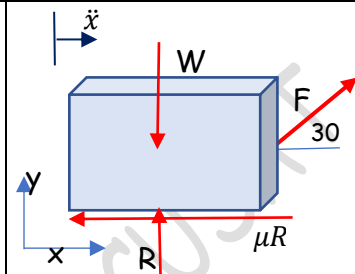
The 50 kg crate shown rests on a horizontal plane for which the coefficient of kinetic friction is 0.3. If the crate is subjected to a towing force $F = 400 \text{ N}$, determine the velocity of the crate in 5 s after starting from rest.

**Kinetics**

Let us draw the free-body diagram.

The forces acting on the body are

1. External force $F = 400 \text{ N}$
2. Weight of the crate $W = 50 \times 9.81 \text{ N}$
3. Normal reaction R
4. Frictional force μR



Considering the fact that the crate moves towards right, the direction of acceleration is assumed to be towards right. As the body does not move in the y-direction, we can write $\Sigma F_y = 0$

$$R - W + F \sin 30 = 0$$

$$R = 50 \text{ kg} \times 9.81 \text{ m/s}^2 - 400 \text{ N} \sin 30 = 290.5 \text{ N}$$

Since the body moves in the x-direction, let us write the equation of motion in the x-direction

$$\Sigma F_x = m\ddot{x}$$

Where ΣF_x produces an acceleration $\ddot{x}$ in its direction, or the force in the direction of acceleration is considered positive and negative otherwise.

$$50\ddot{x} = F \cos 30 - \mu R$$

$$\ddot{x} = \frac{400 \text{ N} \cos 30 - 0.3 \times 290.5 \text{ N}}{50 \text{ kg}} = 5.19 \text{ m/s}^2$$

The acceleration obtained is a constant. This is expected as the forces acting on the system are constant.

Kinematics

The initial velocity, $\dot{x}_0 = 0$

Acceleration, $\ddot{x} = 5.19 \text{ m/s}^2$

Time considered, $t = 5 \text{ s}$

Final velocity, $\dot{x} = ??$

Since the acceleration is constant, we can use the equation $v = u + at$.

$$\dot{x} = \dot{x}_0 + \ddot{x} t = 0 + 5.19 \times 5 = 26 \text{ m/s}$$

Example 2

The driver of a car, travelling along a straight level highway, suddenly applies brakes so that the car slides for 2 s, covering a distance of 10 m before coming to rest. Assuming that during this time the car travelled with uniform deceleration, find the coefficient of friction between the tyres and the pavement.

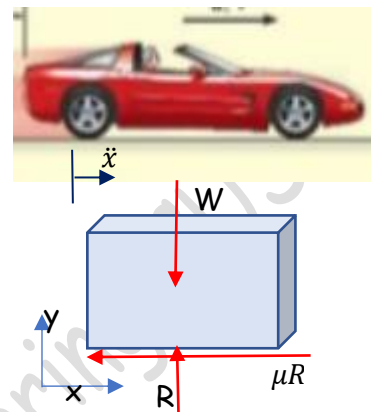
Kinetics

Let us consider the free-body diagram of the car after the brakes were applied.

The forces acting on the car are

1. Weight of the car W
2. Normal reaction R
3. Frictional force μR

As the car undergoes rectilinear translation, it can be considered as a particle. Considering the fact that the car moves towards right, the direction of acceleration is assumed to be towards right.



As the body does not move in the y-direction, we can write $\Sigma F_y = 0$

$$R - W = 0$$

$$R = W$$

Since the body moves in the x-direction, let us write the equation of motion in the x-direction

$$\Sigma F_x = m\ddot{x}$$

Where ΣF_x produces an acceleration $\ddot{x}$ in its direction, or the force in the direction of acceleration is considered positive and negative otherwise.

$$\frac{W}{g}\ddot{x} = -\mu R = -\mu W$$

$$\ddot{x} = -\mu g$$

The acceleration obtained is constant and negative. This is expected as the forces acting on the system are constant and the car was decelerating after the brakes were applied.

Kinematics

The initial velocity, $\dot{x}_0 = ?$

Acceleration, $\ddot{x} = -\mu g \text{ m/s}^2$

Time considered, $t = 2 \text{ s}$

Final velocity, $\dot{x} = 0$

Distance travelled, $x = 10 \text{ m}$

Since $\ddot{x}$ is constant, we can use any two of the equations $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$

$$\dot{x} = \dot{x}_0 + \ddot{x} t$$

$$0 = \dot{x}_0 - \mu g \times 2 \gg \gg \dot{x}_0 = 2\mu g$$

$$\dot{x}^2 = \dot{x}_0^2 + 2\ddot{x} x$$

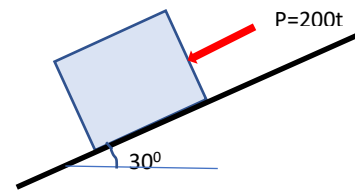
$$0 = (2\mu g)^2 + 10 \times (-2\mu g)$$

$$2\mu g = 10$$

$$\mu = 0.501$$

Example 3

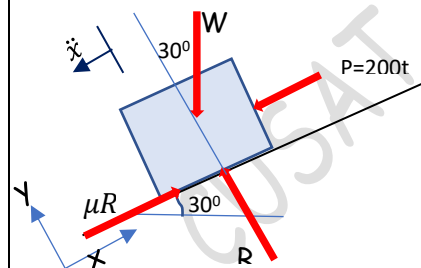
A block of weight 500 N is acted on by a force $P = 200t\text{ N}$, where t is in seconds. Compute its velocity 2 s after P has been applied if the block has an initial velocity $v_0 = 3\text{ m/s}$ down the plane and the coefficient of kinetic friction between the block and the plane is 0.3 .

**Kinetics**

Draw the free-body diagram of the block.

The forces acting on the body are

1. External force $P = 200t\text{ N}$
2. Weight of the crate $W = 500\text{ N}$
3. Normal reaction R
4. Frictional force μR



Let us consider the co-ordinate system as shown. As the block is moving down the plane, the acceleration is considered down the plane.

As the body does not move in the y -direction, we can write $\Sigma F_y = 0$

$$R - W \cos 30 = 0$$

$$R = 500 \cos 30 = 433\text{ N}$$

Since the body moves in the x -direction, let us write the equation of motion in the x -direction

$$\Sigma F_x = \frac{W}{g} \ddot{x}$$

where ΣF_x produces an acceleration $\ddot{x}$ in its direction, or the force in the direction of acceleration is considered positive and negative otherwise.

$$\frac{W}{g} \ddot{x} = P - \mu R + W \sin 30$$

$$\ddot{x} = \frac{9.81}{500} (200t - 0.3 \times 433 + 500 \sin 30) = 3.92t + 2.35$$

Kinematics

The acceleration is not constant, but a function of time because the net force is a function of time. Therefore, we can't use the three formulae we used above.

$$\frac{d\dot{x}}{dt} = 3.92t + 2.35$$

$$d\dot{x} = (3.92t + 2.35)dt$$

Integrating both sides and substituting the initial condition that when $t = 0$, $v_0 = 3\text{ m/s}$, we get

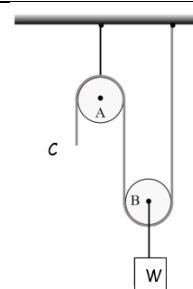
$$\int_{v_0=3\text{ m/s}}^v d\dot{x} = \int_0^2 (3.92t + 2.35)dt$$

$$v - 3 = \left[\frac{3.92t^2}{2} + 2.35t \right]_0^2 = 12.55$$

$$v = 15.55\text{ m/s}$$

Example 4

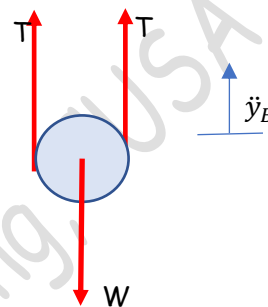
A weight $W = 4450\text{ N}$ is supported in a vertical plane by a string and pulleys arranged as shown. If the free end C of the string is pulled vertically downward with constant acceleration $\ddot{y}_C = 2\text{ m/s}^2$, find the tension in the string neglecting friction and inertia of the pulleys.

**Kinetics**

Consider the free-body diagram of pulley B. We know that as end C is moving down with acceleration $\ddot{y}_C$, pulley B will move up with acceleration $\ddot{y}_B$.

Since the friction in the pulley is neglected, the string tension will remain the same throughout the string. Hence, the forces acting on the system are

1. Weight $W = 4450\text{ N}$
2. Two tensions T and T



There is no force in the horizontal direction. In the y -direction the pulley moves. Hence, we write the equation of motion in the y -direction.

$$\Sigma F_y = \frac{W}{g} \ddot{y}_B$$

$$\frac{W}{g} \ddot{y}_B = 2T - W$$

This is the only equation we can write from the free-body diagram which has two unknowns.

Kinematics

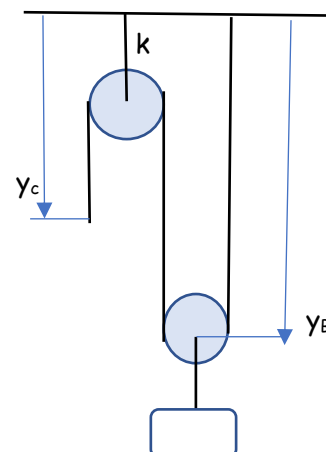
If we can relate the acceleration of end C to the acceleration of pulley B, we can solve the above equation.

Assuming the string is inextensible, we can consider that the length of the string remains constant even if it is moving.

Let us find out the length of the string. As the string is pulled down, the positions of end C and B will change. Assume that y_C and y_B represent the positions of C and B from the fixed support. The length of the string can be written as

$$L = y_C - k + r + y_B - k + r + y_B$$

$$y_C + 2y_B = \text{constant}$$



Differentiating this expression with respect to time twice, we can write,

$$\ddot{y}_C + 2\ddot{y}_B = 0$$

$$\ddot{y}_B = -\frac{\ddot{y}_C}{2} = -1\text{ m/s}^2$$

Negative sign shows that acceleration of B is opposite to that of C (which we have already considered). Hence $\ddot{y}_B = 1\text{ m/s}^2$

$$\frac{W}{g} \ddot{y}_B + W = 2T$$

$$T = \frac{\frac{4450}{9.81} \times 1 + 4450}{2} = 2451\text{ N}$$

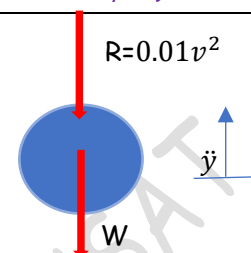
Example 5

A 100 N stone is projected vertically up from the ground with an initial velocity of 50 m/s. Determine the maximum height to which it will travel if the atmospheric resistance is measured as $0.01v^2$ N where v is the speed at any instant measured in m/s. (Assume $g = 10 \text{ m/s}^2$)

Kinetics

Consider the free-body diagram. The forces acting on the ball are

1. Weight $W = 100 \text{ N}$
2. The air resistance $R = 0.01v^2$ in the direction opposite to motion



Write the equation of motion in the vertical direction

$$\Sigma F_y = \frac{W}{g} \ddot{y}$$

$$\frac{W}{g} \ddot{y} = -W - R = -100 - 0.01v^2$$

$$\ddot{y} = -(10 + 0.001v^2)$$

The acceleration is not a constant, but a function of velocity.

Kinematics

Since we need to find the distance, we can use the equation

$$v \frac{dv}{dy} = -(10 + 0.001v^2)$$

$$\frac{-v dv}{(10 + 0.001v^2)} = dy$$

To find the time taken to reach the maximum height

$$\frac{dv}{dt} = -(10 + 0.001v^2)$$

$$\frac{-dv}{(10 + 0.001v^2)} = dt$$

Integrating both sides knowing that $v = 50 \text{ m/s}$ when $t = 0$ and $v = 0$ when the stone reaches the maximum height in time t

$$\int_0^t dt = \int_{50}^0 \frac{-dv}{(10 + 0.001v^2)}$$

$$t = 1000 \int_0^{50} \frac{dv}{(100^2 + v^2)} = 1000 \left[\frac{1}{100} \tan^{-1} \frac{v}{100} \right]_0^{50}$$

$$t = 265.7 \text{ s}$$

Integrating both sides knowing that $v = 50 \text{ m/s}$ when $y = 0$ and $v = 0$ when the stone reaches the maximum height y .

$$\int_{50}^0 \frac{-v dv}{(10 + 0.001v^2)} = \int_0^y dy$$

$$y = \int_0^{50} \frac{v dv}{(10 + 0.001v^2)} = 1000 \int_0^{50} \frac{v dv}{(100^2 + v^2)}$$

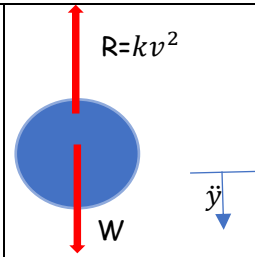
Put $(100^2 + v^2) = u \gg \gg 2v dv = du \gg \gg$ limit changes from $u = 10000$ to $u = 12500$

$$y = 500 \int_{10000}^{12500} \frac{du}{u} = 500(\ln 12500 - \ln 10000) = 111.57 \text{ m}$$

Consider a ball of weight W is dropped from rest at a height h and the air resistance is kv^2 , where k is the drag coefficient. The equation of motion is

$$m\ddot{y} = W - R = mg - kv^2 \ggg \ddot{y} = \left(g - \frac{k}{m}v^2\right)$$

$$\frac{dv}{dt} = \left(g - \frac{k}{m}v^2\right) \lll \frac{dv}{\left(g - \frac{k}{m}v^2\right)} = dt$$



The velocity v of the body after a time t

$$\int_0^t dt = \int_0^v \frac{dv}{\left(g - \frac{k}{m}v^2\right)} = \int_0^v \frac{dv}{\frac{k}{m}\left(\frac{mg}{k} - v^2\right)} = \frac{m}{k} \int_0^v \frac{dv}{\left(\sqrt{\frac{W}{k}}\right)^2 - v^2} = \frac{m}{k} \int_0^v \frac{dv}{a^2 - v^2}; \text{ where } a = \sqrt{\frac{W}{k}}$$

$$t = \frac{m}{k} \times \frac{1}{2a} \ln \left(\frac{a+v}{a-v} \right) \Big|_0^v = \frac{m}{2ak} \ln \left(\frac{a+v}{a-v} \right)$$

$$\ln \left(\frac{a+v}{a-v} \right) = \left(\frac{2ak}{m} \right) t \ggg \frac{a+v}{a-v} = e^{\left(\frac{2ak}{m}\right)t} \ggg a+v = e^{\left(\frac{2ak}{m}\right)t} (a-v)$$

$$a+v = e^{\left(\frac{2ak}{m}\right)t} \times a - e^{\left(\frac{2ak}{m}\right)t} \times v \ggg v \left[1 + e^{\left(\frac{2ak}{m}\right)t} \right] = a \left[e^{\left(\frac{2ak}{m}\right)t} - 1 \right]$$

$$v = \frac{a \left[e^{\left(\frac{2k}{m}\right)t} - 1 \right]}{\left[1 + e^{\left(\frac{2k}{m}\right)t} \right]} = a \times \frac{e^{\left(\frac{2k}{m}\right)t} \left[1 - e^{-\left(\frac{2k}{m}\right)t} \right]}{e^{\left(\frac{2k}{m}\right)t} \left[1 + e^{-\left(\frac{2k}{m}\right)t} \right]} = a \left[\frac{1 - e^{-\left(\frac{2k}{m}\right)t}}{1 + e^{-\left(\frac{2k}{m}\right)t}} \right]$$

As t takes a large value (i.e., $t \rightarrow \infty$), $e^{-\left(\frac{2k}{m}\right)t} \rightarrow e^{-\infty} \rightarrow \frac{1}{e^{\infty}} \rightarrow \frac{1}{\infty} \rightarrow 0$

$$v \rightarrow a = \left(\sqrt{\frac{W}{k}} \right) \text{ m/s (Terminal velocity)}$$

Example 6

A particle of weight W moves rectilinearly under the action of a force $F_x = P \sin \omega t$. Obtain the displacement-time equation assuming that the body starts from rest from the origin.

The differential equation of motion is $\Sigma F_x = \frac{W}{g} \ddot{x}$

$$\frac{W}{g} \ddot{x} = P \sin \omega t \ggg \ddot{x} = \frac{Pg}{W} \sin \omega t$$

Since the force is proportional to time, the acceleration is also proportional to time (not constant)

$$\ddot{x} = \frac{Pg}{W} \sin \omega t \ggg \frac{d\dot{x}}{dt} = \frac{Pg}{W} \sin \omega t \ggg d\dot{x} = \frac{Pg}{W} \sin \omega t dt$$

Integrating both sides knowing that when $t=0$, velocity=0

$$\int_0^v d\dot{x} = \int_0^t \frac{Pg}{W} \sin \omega t dt \ggg v = \left[-\frac{Pg}{W\omega} \cos \omega t \right]_0^t = -\frac{Pg}{W\omega} [\cos \omega t - 1]$$

We know that $v = \frac{dx}{dt}$

$$\frac{dx}{dt} = \frac{Pg}{W\omega} (1 - \cos \omega t)$$

$$dx = \frac{Pg}{W\omega} [1 - \cos \omega t] dt$$

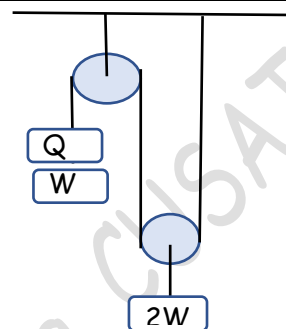
Integrating again with the condition that when $t = 0, x = 0$

$$\int_0^x dx = \frac{Pg}{W\omega} \int_0^t [1 - \cos \omega t] dt \gg x = \frac{Pg}{W\omega} \left[t - \frac{\sin \omega t}{\omega} \right]_0^t = \frac{Pg}{W\omega^2} (\omega t - \sin \omega t)$$

#Exercise 29

29.1

Two weights W and $2W$ are supported in a vertical plane by a string and pulleys arranged as shown. Find the magnitude of the additional weight Q applied on left which will give a downward acceleration of $0.1g$ to the weight W . Neglect friction and inertia of the pulleys

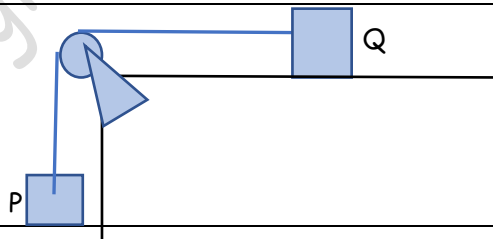


29.2

To determine the coefficient of friction between two materials experimentally, a small block of weight $W = 50 \text{ N}$ is projected with initial velocity of 9 m/s along a horizontal plane covered with the same material. If the block travels a distance of 10 m before coming to rest, what is the coefficient of friction?

29.3

Find the acceleration of the falling weight $P = 45 \text{ N}$ if the coefficient of friction between the block $Q = 53 \text{ N}$ and the plane on which it slides is 0.3 . Neglect inertia and friction in the pulleys

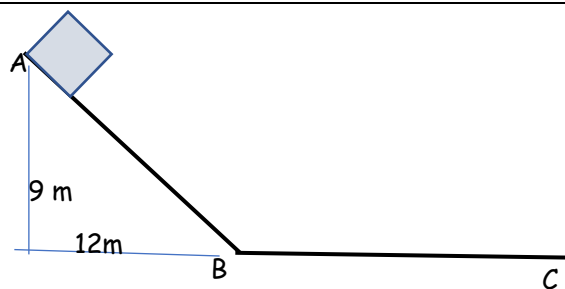


29.4

A small block of weight W rests on an adjustable inclined plane. Friction is such that sliding of the block impends when the inclination of the plane with horizontal is 30° . What acceleration will the block have if the angle of inclination of the plane is 45° ?

29.5

A small block starts from rest from point A and slide down the inclined plane AB as shown. What distance x along the horizontal plane BC will it travel before coming to rest if the coefficient of friction between the planes and the block is 0.3 . Assume that there is no change in the magnitude of velocity at point B



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